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THE ABENDARE TECHNICAL COLLEGE
ICT TECHNICIAN LEVEL 5
CONTINUOUS ASSESSMENT TEST (CAT) 1

TERM 3

84.7

UNIT: COMPUTER SOFTWARE
TIME: 1 HOUR 30 MINUTES
TOTAL MARKS: 50 MARKS

NAME: _____

CDACC REG NO: _____

DATE: _____

INSTRUCTIONS

- ✓ Answer ALL questions.
- ✓ Provide clear and technical explanations where required
- ✓ Use relevant examples where necessary.
- ✓ This paper focuses on software testing and Postman tool usage.

Questions

1. Explain the importance of software testing in ensuring software functionality. (4 Marks)
2. State FOUR reasons why software testing is important in system development. (4 Marks)
3. Differentiate between functional testing and non-functional testing. (4 Marks)
4. State THREE software testing techniques used in verifying software functionality. (3 Marks)
5. Explain TWO advantages of using automated testing over manual testing. (3 Marks)
6. Describe the role of the following software testing tools:
 - a) Test Complete (2 Marks)
 - b) Selenium (2 Marks)
 - c) Postman (2 Marks)
7. Explain what Postman is and its use in software testing. (3 Marks)
8. Describe step-by-step how Postman is used to test an API. (5 Marks)
9. State FOUR HTTP methods used in Postman and briefly explain each. (4 Marks)
10. Explain TWO benefits of API testing using Postman. (3 Marks)
11. Describe the steps involved in testing a new software application before deployment. (4 Marks)
12. State THREE common errors detected during software functionality testing. (3 Marks)
13. Explain TWO challenges faced during software testing and how they can be solved. (3 Marks)
14. Describe how regression testing helps maintain software functionality after changes. (3 Marks)
15. Explain the role of test cases in software functionality testing. (3 Marks)

ii) Reduces maintenance cost ✓
Improves software quality & reliability ✓
Ensures that the software meets the user requirements ✓
Identifies defects & bugs before release ✓

3) Detects errors early ✓
Improves security ✓
Increases user satisfaction ✓
Ensures system requirements are met ✓

3) functional testing

i) Test what the system does ✓
ii) Verifies features and function ✓

Non-functional testing ✓
Tests how well the system performs ✓
Verifies performance, security ✓

4) System testing ✓
integration testing ✓
unit testing ✓

5) faster execution of Repetitive tests ✓
Reduces human error ✓

6) Automates functional & Regression testing for desktop web & mobile application ✓

b) Automates web browser testing & web application validation ✓

c) used to send, create & test API requests & responses ✓

7) Is an API testing tool used to send HTTP requests & inspect responses ✓

8) Open Postman ✓

- Create a new request
- Enter the API endpoint URL
- Select the HTTP method.
- Add header or body if required.
- Send the request and analyse the response.

- 9] DELETE → Removes data.
- GET → Retrieves data.
- POST → Creates new data.
- PUT → Updates existing data.

- 10] Detects API issues before deployment.
- Ensures accurate communication between systems.

facilitates data Exchange and security

- 11] Review requirements.
- Prepare test cases.
- Execute tests & identify defects.
- Fix defects, retest and approve deployment.

- 12] Incorrect outputs
- input validation errors
- Broken links.

- 13] i] Limited testing time
- Prioritize critical test cases & automate tests.

- ii] Incomplete requirements
- Clarify requirements with the stakeholders before testing.

- 14] Ensures that newly added features do not break existing functionality.

- 15] Ensures all requirements are covered.
- Help break test results & defects.
- Provides a structured way of testing features.